RESEARCH PROTOCOL

Metal Biliary Stent versus Plastic Biliary Stent in Patients with Pancreatic Cancer: A Comparative Safety and Effectiveness Study

Version: 0.0.3

Contents

1 List of Abbreviations **2**

2 Responsible Parties **4**

2.1 Investigators 4

2.2 Disclosures 4

3 Abstract **5**

4 Amendments and Updates **6**

5 Rationale and Background **6**

6 Study Objectives **7**

7 Research Methods **7**

7.1 Study Design 7

7.2 Data Sources 7

7.3 Study Population 8

7.4 Exposure Comparators 9

7.5 Outcomes 9
7.6 Analysis 12

8 Sample Size and Study Power **14**

9 Strengths and Limitations **14**
9.1 Strengths 14
9.2 Limitations 15

10 Protection of Human Subjects **15**

11 Management and Reporting of Adverse Events and Adverse Reactions **15**

12 Plans for Disseminating and Communicating Study Results **16**

References **17**

A Exposure Cohort Definitions **18**
A.1 Metal Biliary Stent with Pancreatic Cancer (Target, ATLAS ID: 688) 18
A.2 Plastic Biliary Stent with Pancreatic Cancer (Comparator, ATLAS ID: 689) 22

B Outcome Cohort Definitions **27**
B.1 Acute Cholangitis (Tokyo Guideline 2018, ATLAS Cohort ID: 697) 27
B.2 Moderate Acute Cholangitis (Tokyo Guideline 2018, ATLAS Cohort ID: 719) 28
B.3 Cholangitis - Broad Definition (ATLAS Cohort ID: 663) 30
B.4 All-cause Mortality (ATLAS Cohort ID: 665) 31
B.5 Re-intervention (ATLAS Cohort ID: 696) 31

C Negative Control Concepts **35**

1 List of Abbreviations

CDM	Common data model
OHDSI	Observational Health Data Sciences and Informatics
IRB	Institutional review board
OMOP	Observational Medical Outcomes Partnership
MS	Metal Biliary Stent
PS	Plastic Biliary Stent

2 Responsible Parties

2.1 Investigators

Investigator	Institution/Affiliation
Yiju Park	Department of Biomedical Systems Informatics, Yonsei University College of Medicine, Seoul, Korea
Hee Seung Lee	Department of Internal Medicine, Yonsei University College of Medicine, Seoul, Korea
Seng Chan You*	Department of Biomedical Systems Informatics, Yonsei University College of Medicine, Seoul, Korea

* Principal Investigator

2.2 Disclosures

This study is undertaken within Observational Health Data Sciences and Informatics (OHDSI), an open collaboration.

3 Abstract

Background and Significance: Pancreatic cancer-related biliary obstruction is managed with biliary stent placement via ERCP. Current guidelines recommend metal biliary stents (MS) over plastic biliary stents (PS) for inoperable malignant obstruction; however, real-world comparative evidence on safety and effectiveness remains limited.

Study Aims: To compare the clinical safety and effectiveness of metal versus plastic biliary stents in patients with pancreatic cancer using real-world observational data from the OMOP Common Data Model.

Study Description:

- **Population**: Adult patients (age ≥ 18) with a prior diagnosis of pancreatic cancer who received their first biliary stent insertion, excluding those with a history of cholangiocarcinoma.
- **Comparators**: Target — Metal Biliary Stent (ATLAS Cohort ID: 688); Comparator — Plastic Biliary Stent (ATLAS Cohort ID: 689).
- **Outcomes**: Acute cholangitis (Tokyo Guideline 2018, Cohort ID: 697); Moderate acute cholangitis (Cohort ID: 719); Cholangitis broad definition (Cohort ID: 663); All-cause mortality (Cohort ID: 665); Biliary re-intervention.
- **Design**: New-user active comparator cohort study using 1:1 propensity score matching with large-scale covariates (LASSO logistic regression) and empirical calibration via negative control outcomes.
- **Timeframe**: Follow-up windows of 30, 90, 180, and 365 days from the index date (first stent insertion).

4 Amendments and Updates

Number	Date	Section of study protocol	Amendment or update	Reason
1	2025-03-18	v0.01		Initial draft
2	2026-06-15	v0.02	update	
3	2026-07-24	v0.03	update	feedback

5 Rationale and Background

Pancreatic cancer is an aggressive malignancy with a poor prognosis. At the time of diagnosis, the majority of patients present with biliary obstruction, which causes jaundice, pruritus, and cholangitis, significantly impairing quality of life and overall clinical status. Endoscopic retrograde cholangiopancreatography (ERCP)-guided biliary stent placement is the standard of care for palliating malignant biliary obstruction.

Biliary stents are broadly classified by material into **Metal Biliary Stents (MS)** and **Plastic Biliary Stents (PS)**.

Characteristic	Metal Stent	Plastic Stent
Patency Duration	Long (6–12+ months)	Short (3–6 months)
Cost	High	Low
Exchangeability	Difficult (self-expanding)	Easy
Need for Re-intervention	Low	High

Current domestic and international guidelines (ESGE, ASGE, KSGE) recommend **metal stent use** in patients with inoperable malignant biliary obstruction, based on evidence from randomized controlled trials (RCTs) showing superior patency and lower re-intervention rates compared to plastic stents [[1code],[2code]]. However, whether these guideline recommendations are equally supported by real-world data remains uncertain.

Existing RCTs may not be representative of real-world clinical populations due to strict eligibility criteria. Real-world comparative evidence on safety (cholangitis) and effectiveness (re-intervention, survival) between the two stent types is limited. A large-scale observational study using the OHDSI Common Data Model (CDM) can validate the applicability of current guidelines in real-world settings.

6 Study Objectives

The overall goal of this protocol is conducting comparative safety and effectiveness research to establish evidences for benefits and harms of metal and plastic stents in patients with pancreatic cancer through OHDSI network.

The primary objective is to compare the clinical safety and effectiveness of **metal biliary stents** versus **plastic biliary stents** in patients with pancreatic cancer-related biliary obstruction using real-world data.

Secondary objectives are:

1. To compare the incidence of **acute cholangitis** between the two stent groups
2. To compare the frequency of **biliary re-intervention** between the two stent groups
3. To compare **all-cause mortality** between the two stent groups

7 Research Methods

7.1 Study Design

This study uses `CohortMethod` [[3code]].

This is a **observational study** based on a **new-user active comparator design** using the OHDSI CohortMethod package for population-level effect estimation (PLE). The new-user design mitigates immortal time bias by requiring all patients to be naive to the study exposure at cohort entry [[4code]].

- **Index Date (Day 0):** Date of first biliary stent insertion (metal or plastic)
- **Baseline Period:** 365 days prior to the index date (Day −365 to Day 0)
- **Follow-up Period:** Day 1 through end of observation, death, or end of study period

7.2 Data Sources

The analyses will be performed across a network of observational healthcare databases. All databases have been transformed into the OMOP Common Data Model, version 5.4. The complete specification for OMOP Common Data Model, version 5.4 is available at: https://github.com/OHDSI/CommonDataModel

7.3 Study Population

7.3.1 Target Cohort — Metal Biliary Stent

Definition: Patients with pancreatic cancer whose first-ever biliary stent insertion (of any type) was a metal stent, with no prior biliary stent

Inclusion Criteria:

Criterion	Window	Description
Metal biliary stent insertion	Day 0	First-ever metal biliary stent insertion (Device Exposure)
Age ≥ 18 years	Day 0	
Pancreatic cancer diagnosis	Any time before Day 0	Any recorded diagnosis prior to or on index date
Continuous observation	Day −365 to Day 0	At least 365 days of continuous observation in the database prior to and including the index date

Exclusion Criteria:

Criterion	Window	Description
Cholangiocarcinoma diagnosis	Any time before Day 0	Any prior diagnosis of cholangiocarcinoma
Prior biliary stent (any type)	Any time before Day 0	Any prior metal or plastic biliary stent insertion before index date

7.3.2 Comparator Cohort — Plastic Biliary Stent (ATLAS Cohort ID: 689)

Definition: Patients with pancreatic cancer whose first-ever biliary stent insertion (of any type) was a plastic stent, with no prior biliary stent

Inclusion and exclusion criteria are identical to the target cohort, differing only in the type of stent inserted. Both cohorts additionally require ≥ 365 days of continuous observation prior to the index date to ensure adequate ascertainment of baseline covariates.

7.3.3 Covariate Assessment Windows

Covariate Type	Window
Demographics	Day 0
Short-term baseline conditions and medications	Day −30 to Day -1
Long-term baseline conditions and medications	Day −365 to Day -1

7.4 Exposure Comparators

7.4.1 Target — Metal Biliary Stent

Metal biliary stents are identified in the OMOP CDM `DEVICE_EXPOSURE` table using concept codes representing self-expandable metal stents (SEMS) and related devices (Codeset ID: 2, including descendant concepts for fully covered SEMS).

Only the first-ever metal biliary stent insertion qualifies as the index event, fulfilling the new-user design requirement.

7.4.2 Comparator — Plastic Biliary Stent

Plastic biliary stents are identified in the OMOP CDM `DEVICE_EXPOSURE` table using concept codes representing plastic biliary stent devices.

Only the first-ever plastic biliary stent insertion qualifies as the index event.

7.5 Outcomes

7.5.1 Primary Outcomes

Outcome	Cohort ID	Definition
Acute Cholangitis (Tokyo Guideline 2018)	697	Clinical definition based on Tokyo Guidelines 2018 criteria (see below)
Moderate Acute Cholangitis	719	Moderate severity per Tokyo Guidelines 2018 criteria
Cholangitis (broad definition)	663	Based on diagnosis codes for cholangitis
All-cause Mortality	665	Based on CDM Death table
Re-intervention	696	Based on device or procedure tables

7.5.2 Acute Cholangitis Definition (Tokyo Guideline 2018)

The TG18/TG13 framework [[5code]] defines acute cholangitis using three categories:

- **Suspected diagnosis**: one item in **A** + one item in either **B** or **C**
- **Definite diagnosis**: one item in **A**, one item in **B** and one item in **C**

Operationalization in this study: Category C (imaging) criteria cannot be reliably ascertained from structured OMOP CDM data. Therefore, acute cholangitis in this study is operationalized as **Suspected diagnosis** using Categories A and B only (≥1 item from **A** + ≥1 item from **B**).

Category A — Systemic Inflammation

Criterion	Threshold	OMOP Concept ID
A-1. Fever and/or shaking chills	Body temperature (BT) > 38.0°C	3020891
A-2. WBC count (x 1,000/μL)	<4 or > 10	3000905
A-2. CRP (mg/dL)	≥ 1	3010156

Category B — Cholestasis

Criterion	Threshold	OMOP Concept ID
B-1. Jaundice	Total bilirubin ≥ 2.0 mg/dL	3024128
B-2. Liver enzymes	ALP, γGTP, AST, and ALT > 1.5× upper limit of normal	—

Category C — Imaging *(not ascertainable from CDM; excluded from operationalization)*

Criterion	Description
Biliary dilatation Evidence of etiology on imaging(stricture, stone, stent etc.)	Stricture, stone, stent, or other obstructing cause on imaging

7.5.2.1 Severity Grading (TG18) Grade I (Mild) and Grade II (Moderate) severity criteria were each operationalized as a separate outcome cohort, defined as Acute Cholangitis (Cohort ID: 697) and Moderate Acute Cholangitis (Cohort ID: 719)

Grade I (Mild): Does not meet Grade II or III criteria.

Grade II (Moderate): ≥2 of the following 5 criteria:

Criterion	Threshold
WBC	> 12,000 or < 4,000/mm³
Fever	BT ≥ 39°C
Age	≥ 75 years
Total bilirubin	≥ 5 mg/dL
Albumin	< 0.7 × lower limit of normal

Grade III (Severe): ≥1 of the following organ/system dysfunctions:

System	Criterion
Cardiovascular	Hypotension requiring dopamine ≥ 5 μg/kg/min or any dose of norepinephrine
Neurological	Disturbance of consciousness
Respiratory	PaO□/FiO□ < 300
Renal	Oliguria or serum creatinine > 2.0 mg/dL
Hepatic	PT-INR > 1.5
Hematological	Platelet count < 100,000/mm³

7.5.3 Biliary Re-intervention

Definition: A biliary re-intervention event is defined as any procedure code indicating biliary stent replacement, reinsertion, or repair occurring after Day 7 from the index date (first stent insertion).

Identification: Biliary re-intervention is identified using (1) the same concept sets as the exposure cohorts (metal or plastic biliary stent device exposure codes), and (2) relevant concepts from the Procedure domain, including repeat ERCP, percutaneous transhepatic biliary drainage (PTBD) insertion or exchange, and other biliary drainage procedures that do not involve in device exposure. An event is counted as a re-intervention if it occurs more than 7 days after the index date, regardless of the stent type used at re-intervention.

Rationale for Day 7 washout: Events within the first 7 days post-index are excluded to avoid misclassifying planned staged procedures or early technical corrections as re-interventions.

7.6 Analysis

7.6.1 Covariates

Propensity scores are estimated using large-scale covariate sets from the `FeatureExtraction` package. Concept codes directly related to stent insertion are excluded from the covariate set (`excludedCovariateConceptIds`).

Included Covariates:

Domain	Detail	Window
Demographics	Age group, sex, race, calendar month	Day 0
Conditions	Condition Group Era	Short-term (−30 to -1), Long-term (−365 to -1)
Medications	Drug Group Era	Short-term (−30 to -1), Long-term (−365 to -1)
Measurements	Measurement values	Short-term (−30 to -1), Long-term (−365 to -1)
Measurement ranges	Measurement Range Group	Short-term (−30 to -1), Long-term (−365 to -1)
Observations	Observation	Short-term (−30 to -1), Long-term (−365 to -1))
Composite indices	Charlson Comorbidity Index	Day 0

7.6.2 Propensity Score Matching

- **Model:** LASSO Logistic Regression (via `Cyclops` package)
- **Matching:** 1:1 nearest-neighbor matching without replacement
- **Caliper:** 0.2 (standardized logit scale)
- **Common Support:** Preference score-based trimming (PS trimming)

7.6.3 Balance Assessment

Covariate balance before and after matching is assessed using standardized mean differences (SMD). Target threshold: all covariates SMD ≤ 0.1 (or ≤ 0.25).

7.6.4 Follow-up Windows

Analysis ID	Description	Follow-up Period
1	ITT 1:1 PS Matching	90 days
2	ITT 1:1 PS Matching	30 days
3	ITT 1:1 PS Matching	180 days
4	ITT 1:1 PS Matching	365 days

All analyses apply the intent-to-treat (ITT) principle.

7.6.5 Outcome Model

Primary Analysis

- **Model:** Cause-specific Cox Proportional Hazard Model
- **Effect Estimate:** Hazard Ratio (HR) with 95% Confidence Interval
- **Censoring:** All-cause death is treated as a non-informative censoring event for the cholangitis and re-intervention outcomes
- **Primary Reporting Timepoint:** 365 days from index date (Analysis ID 4)
- **Survival Visualization (All-cause Mortality only):** Kaplan-Meier survival curves

Sensitivity Analysis — Competing Risk of Death

Given the high mortality rate associated with pancreatic cancer, all-cause death constitutes a competing risk for the acute cholangitis and biliary re-intervention outcomes. Standard Cox/KM approaches that treat death as non-informative censoring may overestimate the cumulative incidence of these outcomes. To address this, the following sensitivity analyses are conducted for the cholangitis and re-intervention outcomes only:

- **Model:** Fine-Gray subdistribution hazard model, with all-cause death specified as a competing risk
- **Effect Estimate:** Subdistribution Hazard Ratio (sHR) with 95% Confidence Interval
- **Visualization:** Cumulative Incidence Function (CIF), accounting for the competing risk of death, presented alongside the cause-specific Cox model results

All-cause mortality is not subject to competing risk considerations and is therefore analyzed using the cause-specific Cox model and Kaplan-Meier visualization only, as specified above.

7.6.6 Empirical Calibration

Negative control outcomes are used to assess and correct for residual confounding and systematic bias using the `EmpiricalCalibration` package [[6code]].

- **Negative Controls:** Outcomes with no known biological association with either stent type (see Negative Control Concepts)
- **Calibration Procedure:**
  - Pre-calibration: Assess proportion of negative controls with HR = 1 within 95% CI
  - Post-calibration: Estimate and correct for systematic bias; derive calibrated effect estimates

8 Sample Size and Study Power

This is a retrospective cohort study using existing electronic health records. Sample sizes are determined by available data rather than pre-specified power calculations.

Available sample sizes prior to matching:

Group	N
Metal biliary stent (Target)	1,273
Plastic biliary stent (Comparator)	870

Power calculations will be conducted based on observed event rates for each outcome. Adequate power is anticipated for the primary outcomes (acute cholangitis and re-intervention) given expected event rates. Statistical power for all-cause mortality analyses may be limited due to the overall poor prognosis of pancreatic cancer.

9 Strengths and Limitations

9.1 Strengths

1. **Real-world data:** Reflects actual clinical populations including patients excluded from RCTs due to strict eligibility criteria
2. **Large-scale propensity score matching:** Hundreds of covariates used for robust confounding control

3. **Empirical calibration:** Quantitative assessment and correction of residual confounding and systematic bias using negative control outcomes
4. **OHDSI standard methodology:** Reproducible and transparent analytical framework
5. **Multiple follow-up windows:** Evaluation at 30, 90, 180, and 365 days covers diverse clinically relevant timepoints

9.2 Limitations

1. **Observational study limitations:** Absence of randomization means unmeasured confounding cannot be fully excluded
2. **Single database:** Generalizability of results may be limited
3. **Outcome misclassification — acute cholangitis:** The TG18 definite diagnosis requires imaging evidence of biliary obstruction or dilatation (Category C); however, structured imaging findings are not reliably captured in the OMOP CDM. Consequently, acute cholangitis is operationalized using a **suspected diagnosis** criterion (Categories A + B only), which may result in outcome misclassification. Cases meeting only laboratory criteria without imaging confirmation may represent other etiologies, potentially inflating the observed incidence of cholangitis. 4.**Competing risk of death:** The standard Cox proportional hazards and Kaplan-Meier approach assumes death is a non-informative censoring event. Given the high mortality rate in pancreatic cancer patients, this assumption may be violated, potentially overestimating the cumulative incidence of cholangitis and re-intervention. To mitigate this limitation, a sensitivity analysis using the Fine-Gray subdistribution hazard model, treating all-cause death as a competing risk, is conducted for these outcomes.

10 Protection of Human Subjects

The study is using only de-identified data. Confidentiality of patient records will be maintained at all times. All study reports will contain aggregate data only and will not identify individual patients or physicians.

11 Management and Reporting of Adverse Events and Adverse Reactions

This is a non-interventional, retrospective observational study using existing electronic health records. No investigational products are administered and no direct patient interventions are performed. Accordingly, formal adverse event monitoring and reporting

procedures applicable to interventional studies are not required. Any incidental findings of clinical significance identified during data analysis will be handled in accordance with applicable institutional policies.

12 Plans for Disseminating and Communicating Study Results

The study results will be posted on the OHDSI website after completion of the study. At least one paper describing the study and its results will be written and submitted for publication to a peer-reviewed scientific journal.

References

1 Dumonceau J-M, Tringali A, Papanikolaou IS, *et al.* Endoscopic biliary stenting: Indications, choice of stents, and results: European Society of Gastrointestinal Endoscopy (ESGE) Clinical Guideline – updated October 2017. *Endoscopy* 2018;**50**:910–30. doi:10.1055/a-0659-9864

2 Almadi MA, Barkun A, Martel M. No benefit of covered vs uncovered self-expandable metal stents in patients with malignant distal biliary obstruction: A meta-analysis. *Clinical Gastroenterology and Hepatology* 2013;**11**:27–37. doi:10.1016/j.cgh.2012.10.019

3 Schuemie MJ, Reps JM, Black A, *et al.* Health-analytics data to evidence suite (HADES): Open-source software for observational research. *Studies in Health Technology and Informatics* 2024;**310**:966–70. doi:10.3233/SHTI231108

4 Suissa S. Immortal time bias in pharmaco-epidemiology. *American Journal of Epidemiology* 2008;**167**:492–9. doi:10.1093/aje/kwm324

5 Kiriyama S, Kozaka K, Takada T, *et al.* Tokyo guidelines 2018: Diagnostic criteria and severity grading of acute cholangitis (with videos). *Journal of Hepato-Biliary-Pancreatic Sciences* 2018;**25**:17–30. doi:10.1002/jhbp.512

6 Schuemie MJ, Ryan PB, Hripcsak G, *et al.* Empirical confidence interval calibration for population-level effect estimation studies in observational healthcare data. *Proceedings of the National Academy of Sciences of the United States of America* 2018;**115**:2571–7. doi:10.1073/pnas.1708282114

Appendix

A Exposure Cohort Definitions

A.1 Metal Biliary Stent with Pancreatic Cancer (Target, ATLAS ID: 688)

A.1.1 Cohort Entry Events

People with continuous observation of 365 days before event may enter the cohort when observing any of the following:

1. device exposures of '[yiju] Metal Biliary Stent'.

Limit cohort entry events to the earliest event per person.

A.1.2 Additional Inclusion Criteria

I. Age of more than 18 years Entry events with the following event criteria: who are > 18 years old. #### II. Patient with Pancreatic cancer (all-0d) {-}

Entry events having at least 1 condition occurrence of '[yiju] Cancer of pancreas', starting anytime on or before cohort entry start date; allow events outside observation period.

III. Patient without Bile duct cancer (all-0d) Entry events having no condition occurrences of '[yiju] Cancer of bile duct', starting anytime on or before cohort entry start date; allow events outside observation period.

IV. Patient without Metal/Plastic biliary Stent before (all-1d) Entry events having no device exposures of '[yiju] Biliary Stent', starting anytime prior to cohort entry start date; allow events outside observation period.

A.1.3 Cohort Exit

The person also exists the cohort at the end of continuous observation.

A.1.4 Cohort Eras

Remaining events will be combined into cohort eras if they are within 0 days of each other.

A.1.5 Concept: [yiju] Cancer of pancreas

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
4180793	Malignant tumor of pancreas	363418001	SNOMED	NO	NO	NO
37204852	Acinar cell carcinoma of pancreas	783771003	SNOMED	NO	NO	NO
45763891	Adenocarcinoma of pancreas	700423003	SNOMED	NO	NO	NO
4115272	Carcinoma of ampulla of Vater	254609000	SNOMED	NO	NO	NO
4181331	Carcinoma of body of pancreas	363368005	SNOMED	NO	NO	NO
4110585	Carcinoma of endocrine pancreas	254612002	SNOMED	NO	NO	NO
4209933	Carcinoma of head of pancreas	326072005	SNOMED	NO	NO	NO
4157459	Carcinoma of pancreas	372142002	SNOMED	NO	NO	NO
4178960	Carcinoma of tail of pancreas	363369002	SNOMED	NO	NO	NO
4340498	Cystadenocarcinoma of pancreas	235966007	SNOMED	NO	NO	NO
37395837	Familial malignant neoplasm of pancreas	715414009	SNOMED	NO	NO	NO
36683250	Invasive intraductal papillary-mucinous carcinoma of pancreas	780821007	SNOMED	NO	NO	NO
4201481	Local recurrence of malignant tumor of pancreas	314964006	SNOMED	NO	NO	NO
765387	Malignant carcinoid tumor of pancreas	143401000119106	SNOMED	NO	NO	NO
37017016	Malignant insulinoma	713189001	SNOMED	NO	NO	NO
764972	Malignant melanoma metastatic to pancreas	97051000119105	SNOMED	NO	NO	NO
4092074	Malignant neoplasm of ectopic pancreatic tissue	187798008	SNOMED	NO	NO	NO
4092069	Malignant neoplasm of sphincter of Oddi	187786003	SNOMED	NO	NO	NO
4178966	Malignant tumor of ampulla of Vater	363417006	SNOMED	NO	NO	NO
4092072	Malignant tumor of body of pancreas	187791002	SNOMED	NO	NO	NO
4112734	Malignant tumor of endocrine pancreas	254611009	SNOMED	NO	NO	NO
4111024	Malignant tumor of exocrine pancreas	255088001	SNOMED	NO	NO	NO
4178967	Malignant tumor of head of pancreas	363419009	SNOMED	NO	NO	NO
4095437	Malignant tumor of Islets of Langerhans	187794005	SNOMED	NO	NO	NO
4180793	Malignant tumor of pancreas	363418001	SNOMED	NO	NO	NO
4094866	Malignant tumor of pancreatic duct	187793004	SNOMED	NO	NO	NO
4095436	Malignant tumor of tail of pancreas	187792009	SNOMED	NO	NO	NO
4112774	Metastasis to pancreas of unknown primary	285614004	SNOMED	NO	NO	NO
3655588	Mixed ductal-neuroendocrine carcinoma of pancreas	866052009	SNOMED	NO	NO	NO
37206235	Mucinous cystadenocarcinoma of pancreas	785879009	SNOMED	NO	NO	NO
3655584	Mucinous cystic neoplasm with invasive carcinoma of pancreas	866048009	SNOMED	NO	NO	NO
192261	Overlapping malignant neoplasm of pancreas	109848009	SNOMED	NO	NO	NO
37311469	Pancreatic ductal adenocarcinoma	792907004	SNOMED	NO	NO	NO
36715928	Primary adenocarcinoma of ampulla of Vater	721718003	SNOMED	NO	NO	NO
36713362	Primary adenocarcinoma of body of pancreas	681621000119105	SNOMED	NO	NO	NO
36713363	Primary adenocarcinoma of head of pancreas	681721000119103	SNOMED	NO	NO	NO
42872399	Primary adenocarcinoma of pancreas	1651000119109	SNOMED	NO	NO	NO
441225	Primary malignant neoplasm of ampulla of Vater	371967001	SNOMED	NO	NO	NO
434293	Primary malignant neoplasm of body of pancreas	93715005	SNOMED	NO	NO	NO

440649	Primary malignant neoplasm of head of pancreas	372119009	SNOMED	NO	NO	NO
25486	Primary malignant neoplasm of islets of Langerhans	93843007	SNOMED	NO	NO	NO
199754	Primary malignant neoplasm of pancreas	372003004	SNOMED	NO	NO	NO
433423	Primary malignant neoplasm of pancreatic duct	93939009	SNOMED	NO	NO	NO
432843	Primary malignant neoplasm of tail of pancreas	94082003	SNOMED	NO	NO	NO
37116589	Primary malignant neuroendocrine neoplasm of ampulla of Vater	733351008	SNOMED	NO	NO	NO
42536743	Primary malignant neuroendocrine neoplasm of pancreas	735735001	SNOMED	NO	NO	NO
4311634	Secondary malignant neoplasm of ampulla of Vater	94164003	SNOMED	NO	NO	NO
4246368	Secondary malignant neoplasm of body of pancreas	94212002	SNOMED	NO	NO	NO
4312017	Secondary malignant neoplasm of head of pancreas	94325008	SNOMED	NO	NO	NO
4248063	Secondary malignant neoplasm of islets of Langerhans	94354002	SNOMED	NO	NO	NO
4248192	Secondary malignant neoplasm of pancreas	94459006	SNOMED	NO	NO	NO
4315550	Secondary malignant neoplasm of pancreatic duct	94460001	SNOMED	NO	NO	NO
4311347	Secondary malignant neoplasm of tail of pancreas	94618007	SNOMED	NO	NO	NO
37204808	Serous cystadenocarcinoma of pancreas	783706007	SNOMED	NO	NO	NO
37204187	Solid pseudopapillary carcinoma of pancreas	782697005	SNOMED	NO	NO	NO
36674768	Squamous cell carcinoma of exocrine pancreas	770602005	SNOMED	NO	NO	NO

A.1.6 Concept: [yiju] Cancer of bile duct

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
4181345	Malignant tumor of biliary tract	363415003	SNOMED	NO	NO	NO
35623402	Adenocarcinoma of gallbladder and extrahepatic biliary tract	765741003	SNOMED	NO	NO	NO
37204022	Adenocarcinoma of liver and intrahepatic biliary tract	787091002	SNOMED	NO	NO	NO
4116242	Carcinoma common bile duct	255086002	SNOMED	NO	NO	NO
4115272	Carcinoma of ampulla of Vater	254609000	SNOMED	NO	NO	NO
4162877	Carcinoma of extrahepatic bile duct	372101000	SNOMED	NO	NO	NO
4162998	Carcinoma of gallbladder	372140005	SNOMED	NO	NO	NO
4208660	Cholangiocarcinoma of biliary tract	312104005	SNOMED	NO	NO	NO
4001664	Intrahepatic bile duct carcinoma	109842005	SNOMED	NO	NO	NO
4098954	Klatskin's tumor	253017000	SNOMED	NO	NO	NO
4196255	Local recurrence of malignant tumor of gallbladder	314962005	SNOMED	NO	NO	NO
3655605	Malignant mesenchymal neoplasm of bile duct	866070007	SNOMED	NO	NO	NO
3655606	Malignant mesenchymal neoplasm of gallbladder	866071006	SNOMED	NO	NO	NO
4095434	Malignant neoplasm of hepatic duct	187784000	SNOMED	NO	NO	NO
4089662	Malignant neoplasm of interlobular bile ducts	187773007	SNOMED	NO	NO	NO
4089663	Malignant neoplasm of intrahepatic canaliculi	187776004	SNOMED	NO	NO	NO
4094865	Malignant neoplasm of intrahepatic gall duct	187777008	SNOMED	NO	NO	NO
4092069	Malignant neoplasm of sphincter of Oddi	187786003	SNOMED	NO	NO	NO

4116243	Malignant polyp of biliary tract	255087006	SNOMED	NO	NO	NO
4178966	Malignant tumor of ampulla of Vater	363417006	SNOMED	NO	NO	NO
4177105	Malignant tumor of extrahepatic bile duct	363416002	SNOMED	NO	NO	NO
4181327	Malignant tumor of gallbladder	363353009	SNOMED	NO	NO	NO
198695	Overlapping malignant neoplasm of biliary tract	109847004	SNOMED	NO	NO	NO
3655520	Perihilar cholangiocarcinoma	865951006	SNOMED	NO	NO	NO
36715928	Primary adenocarcinoma of ampulla of Vater	721718003	SNOMED	NO	NO	NO
37116592	Primary adenocarcinoma of biliary tract	733354000	SNOMED	NO	NO	NO
36717173	Primary adenocarcinoma of common bile duct	721559007	SNOMED	NO	NO	NO
36715798	Primary adenocarcinoma of cystic duct	721558004	SNOMED	NO	NO	NO
37018657	Primary adenocarcinoma of gallbladder	9541000119105	SNOMED	NO	NO	NO
36715927	Primary cholangiocarcinoma of intrahepatic biliary tract	721716004	SNOMED	NO	NO	NO
441225	Primary malignant neoplasm of ampulla of Vater	371967001	SNOMED	NO	NO	NO
194589	Primary malignant neoplasm of biliary tract	371970002	SNOMED	NO	NO	NO
4246034	Primary malignant neoplasm of common bile duct	93763008	SNOMED	NO	NO	NO
4247824	Primary malignant neoplasm of cystic duct	93770008	SNOMED	NO	NO	NO
40492416	Primary malignant neoplasm of distal bile duct	447416008	SNOMED	NO	NO	NO
40486896	Primary malignant neoplasm of extrahepatic bile duct	446189008	SNOMED	NO	NO	NO
197806	Primary malignant neoplasm of gallbladder	372139008	SNOMED	NO	NO	NO
40490929	Primary malignant neoplasm of intrahepatic bile duct	447109003	SNOMED	NO	NO	NO
40489795	Primary malignant neoplasm of perihilar bile duct	446807009	SNOMED	NO	NO	NO
37116589	Primary malignant neuroendocrine neoplasm of ampulla of Vater	733351008	SNOMED	NO	NO	NO
37116591	Primary malignant neuroendocrine neoplasm of biliary tract	733353006	SNOMED	NO	NO	NO
37116586	Primary malignant neuroendocrine neoplasm of cystic duct	733348001	SNOMED	NO	NO	NO
37116588	Primary malignant neuroendocrine neoplasm of distal bile duct	733350009	SNOMED	NO	NO	NO
42538835	Primary malignant neuroendocrine neoplasm of perihilar bile duct	762665002	SNOMED	NO	NO	NO
37116590	Primary mucinous cystic neoplasm with associated invasive carcinoma of biliary tract	733352001	SNOMED	NO	NO	NO
37118669	Primary mucinous cystic neoplasm with associated invasive carcinoma of cystic duct	733347006	SNOMED	NO	NO	NO
37116587	Primary mucinous cystic neoplasm with associated invasive carcinoma of distal bile duct	733349009	SNOMED	NO	NO	NO
37116585	Primary mucinous cystic neoplasm with associated invasive carcinoma of perihilar bile duct	733346002	SNOMED	NO	NO	NO
4311634	Secondary malignant neoplasm of ampulla of Vater	94164003	SNOMED	NO	NO	NO
4311638	Secondary malignant neoplasm of biliary tract	94185003	SNOMED	NO	NO	NO
4247946	Secondary malignant neoplasm of common bile duct	94262007	SNOMED	NO	NO	NO
4247947	Secondary malignant neoplasm of cystic duct	94270002	SNOMED	NO	NO	NO
4312301	Secondary malignant neoplasm of extrahepatic bile ducts	94291005	SNOMED	NO	NO	NO
4313079	Secondary malignant neoplasm of gallbladder	94312000	SNOMED	NO	NO	NO

4312024	Secondary malignant neoplasm of intrahepatic bile ducts	94349006	SNOMED	NO	NO	NO
35624313	Squamous cell carcinoma of gallbladder and extrahepatic biliary tract	766978002	SNOMED	NO	NO	NO
36674831	Squamous cell carcinoma of liver and intrahepatic biliary tract	770684008	SNOMED	NO	NO	NO
36674832	Undifferentiated carcinoma of liver and intrahepatic biliary tract	770685009	SNOMED	NO	NO	NO

A.1.7 Concept: [yiju] Metal Biliary Stent

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
45764662	Bare metal biliary stent	701398005	SNOMED	NO	YES	NO
42087127	ADVANIX BILIARY STENT	J5202031	EDI	NO	NO	NO
42088559	BILIARY STENT	J5202105	EDI	NO	NO	NO
42089079	C-FLEX PIGTAIL BILIARY STENT	J5202021	EDI	NO	NO	NO
42090449	COTTON-LEUNG BILIARY STENT	J5202282	EDI	NO	NO	NO
42090450	COTTON-LEUNG BILIARY STENT	J5202382	EDI	NO	NO	NO
42092740	FLEXIMA BILIARY STENT SYSTEM	J5202011	EDI	NO	NO	NO
42098363	PERCUFLEX AMSTERDAM BILIARY STENT	J5202001	EDI	NO	NO	NO
42100653	SOLUS BILIARY STENT	J5203182	EDI	NO	NO	NO
42100842	ST-2 SOEHENDRA TANNENBAUM BILIARY STENT	J5203082	EDI	NO	NO	NO
42103466	ZIMMON BILIARY STENT	J5202482	EDI	NO	NO	NO
42103467	ZIMMON BILIARY STENT	J5202582	EDI	NO	NO	NO

A.1.8 Concept: [yiju] Biliary Stent

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
4234567	Biliary stent	360123005	SNOMED	NO	YES	NO

A.2 Plastic Biliary Stent with Pancreatic Cancer (Comparator, ATLAS ID: 689)

A.2.1 Cohort Entry Events

People with continuous observation of 365 days before event may enter the cohort when observing any of the following:

1. device exposures of '[yiju] Plastic Biliary Stent'.

Limit cohort entry events to the earliest event per person.

A.2.2 Additional Inclusion Criteria

I. Age of more than 18 years Entry events with the following event criteria: who are > 18 years old. #### II. Patient with Pancreatic cancer (all-0d) {-}

Entry events having at least 1 condition occurrence of '[yiju] Cancer of pancreas', starting anytime on or before cohort entry start date; allow events outside observation period.

III. Patient without Bile duct cancer (all-0d) Entry events having no condition occurrences of '[yiju] Cancer of biliary stent', starting anytime on or before cohort entry start date; allow events outside observation period.

IV. Patient without Metal/Plastic biliary stent before (all-1d) Entry events having no device exposures of '[yiju] Biliary stent', starting anytime prior to cohort entry start date; allow events outside observation period.

A.2.3 Cohort Exit

The person also exists the cohort at the end of continuous observation.

A.2.4 Cohort Eras

Remaining events will be combined into cohort eras if they are within 0 days of each other.

A.2.5 Concept: [yiju] Cancer of pancreas

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
4180793	Malignant tumor of pancreas	363418001	SNOMED	NO	NO	NO
37204852	Acinar cell carcinoma of pancreas	783771003	SNOMED	NO	NO	NO
45763891	Adenocarcinoma of pancreas	700423003	SNOMED	NO	NO	NO
4115272	Carcinoma of ampulla of Vater	254609000	SNOMED	NO	NO	NO
4181331	Carcinoma of body of pancreas	363368005	SNOMED	NO	NO	NO
4110585	Carcinoma of endocrine pancreas	254612002	SNOMED	NO	NO	NO
4209933	Carcinoma of head of pancreas	326072005	SNOMED	NO	NO	NO
4157459	Carcinoma of pancreas	372142002	SNOMED	NO	NO	NO
4178960	Carcinoma of tail of pancreas	363369002	SNOMED	NO	NO	NO
4340498	Cystadenocarcinoma of pancreas	235966007	SNOMED	NO	NO	NO
37395837	Familial malignant neoplasm of pancreas	715414009	SNOMED	NO	NO	NO
36683250	Invasive intraductal papillary-mucinous carcinoma of pancreas	780821007	SNOMED	NO	NO	NO
4201481	Local recurrence of malignant tumor of pancreas	314964006	SNOMED	NO	NO	NO
765387	Malignant carcinoid tumor of pancreas	143401000119106	SNOMED	NO	NO	NO

37017016	Malignant insulinoma	713189001	SNOMED	NO	NO	NO
764972	Malignant melanoma metastatic to pancreas	97051000119105	SNOMED	NO	NO	NO
4092074	Malignant neoplasm of ectopic pancreatic tissue	187798008	SNOMED	NO	NO	NO
4092069	Malignant neoplasm of sphincter of Oddi	187786003	SNOMED	NO	NO	NO
4178966	Malignant tumor of ampulla of Vater	363417006	SNOMED	NO	NO	NO
4092072	Malignant tumor of body of pancreas	187791002	SNOMED	NO	NO	NO
4112734	Malignant tumor of endocrine pancreas	254611009	SNOMED	NO	NO	NO
4111024	Malignant tumor of exocrine pancreas	255088001	SNOMED	NO	NO	NO
4178967	Malignant tumor of head of pancreas	363419009	SNOMED	NO	NO	NO
4095437	Malignant tumor of Islets of Langerhans	187794005	SNOMED	NO	NO	NO
4180793	Malignant tumor of pancreas	363418001	SNOMED	NO	NO	NO
4094866	Malignant tumor of pancreatic duct	187793004	SNOMED	NO	NO	NO
4095436	Malignant tumor of tail of pancreas	187792009	SNOMED	NO	NO	NO
4112774	Metastasis to pancreas of unknown primary	285614004	SNOMED	NO	NO	NO
3655588	Mixed ductal-neuroendocrine carcinoma of pancreas	866052009	SNOMED	NO	NO	NO
37206235	Mucinous cystadenocarcinoma of pancreas	785879009	SNOMED	NO	NO	NO
3655584	Mucinous cystic neoplasm with invasive carcinoma of pancreas	866048009	SNOMED	NO	NO	NO
192261	Overlapping malignant neoplasm of pancreas	109848009	SNOMED	NO	NO	NO
37311469	Pancreatic ductal adenocarcinoma	792907004	SNOMED	NO	NO	NO
36715928	Primary adenocarcinoma of ampulla of Vater	721718003	SNOMED	NO	NO	NO
36713362	Primary adenocarcinoma of body of pancreas	681621000119105	SNOMED	NO	NO	NO
36713363	Primary adenocarcinoma of head of pancreas	681721000119103	SNOMED	NO	NO	NO
42872399	Primary adenocarcinoma of pancreas	1651000119109	SNOMED	NO	NO	NO
441225	Primary malignant neoplasm of ampulla of Vater	371967001	SNOMED	NO	NO	NO
434293	Primary malignant neoplasm of body of pancreas	93715005	SNOMED	NO	NO	NO
440649	Primary malignant neoplasm of head of pancreas	372119009	SNOMED	NO	NO	NO
25486	Primary malignant neoplasm of islets of Langerhans	93843007	SNOMED	NO	NO	NO
199754	Primary malignant neoplasm of pancreas	372003004	SNOMED	NO	NO	NO
433423	Primary malignant neoplasm of pancreatic duct	93939009	SNOMED	NO	NO	NO
432843	Primary malignant neoplasm of tail of pancreas	94082003	SNOMED	NO	NO	NO
37116589	Primary malignant neuroendocrine neoplasm of ampulla of Vater	733351008	SNOMED	NO	NO	NO
42536743	Primary malignant neuroendocrine neoplasm of pancreas	735735001	SNOMED	NO	NO	NO
4311634	Secondary malignant neoplasm of ampulla of Vater	94164003	SNOMED	NO	NO	NO
4246368	Secondary malignant neoplasm of body of pancreas	94212002	SNOMED	NO	NO	NO
4312017	Secondary malignant neoplasm of head of pancreas	94325008	SNOMED	NO	NO	NO
4248063	Secondary malignant neoplasm of islets of Langerhans	94354002	SNOMED	NO	NO	NO
4248192	Secondary malignant neoplasm of pancreas	94459006	SNOMED	NO	NO	NO
4315550	Secondary malignant neoplasm of pancreatic duct	94460001	SNOMED	NO	NO	NO
4311347	Secondary malignant neoplasm of tail of pancreas	94618007	SNOMED	NO	NO	NO
37204808	Serous cystadenocarcinoma of pancreas	783706007	SNOMED	NO	NO	NO
37204187	Solid pseudopapillary carcinoma of pancreas	782697005	SNOMED	NO	NO	NO

36674768	Squamous cell carcinoma of exocrine pancreas	770602005	SNOMED	NO	NO	NO

A.2.6 Concept: [yiju] Cancer of biliary stent

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
4181345	Malignant tumor of biliary tract	363415003	SNOMED	NO	NO	NO
35623402	Adenocarcinoma of gallbladder and extrahepatic biliary tract	765741003	SNOMED	NO	NO	NO
37204022	Adenocarcinoma of liver and intrahepatic biliary tract	787091002	SNOMED	NO	NO	NO
4116242	Carcinoma common bile duct	255086002	SNOMED	NO	NO	NO
4115272	Carcinoma of ampulla of Vater	254609000	SNOMED	NO	NO	NO
4162877	Carcinoma of extrahepatic bile duct	372101000	SNOMED	NO	NO	NO
4162998	Carcinoma of gallbladder	372140005	SNOMED	NO	NO	NO
4208660	Cholangiocarcinoma of biliary tract	312104005	SNOMED	NO	NO	NO
4001664	Intrahepatic bile duct carcinoma	109842005	SNOMED	NO	NO	NO
4098954	Klatskin's tumor	253017000	SNOMED	NO	NO	NO
4196255	Local recurrence of malignant tumor of gallbladder	314962005	SNOMED	NO	NO	NO
3655605	Malignant mesenchymal neoplasm of bile duct	866070007	SNOMED	NO	NO	NO
3655606	Malignant mesenchymal neoplasm of gallbladder	866071006	SNOMED	NO	NO	NO
4095434	Malignant neoplasm of hepatic duct	187784000	SNOMED	NO	NO	NO
4089662	Malignant neoplasm of interlobular bile ducts	187773007	SNOMED	NO	NO	NO
4089663	Malignant neoplasm of intrahepatic canaliculi	187776004	SNOMED	NO	NO	NO
4094865	Malignant neoplasm of intrahepatic gall duct	187777008	SNOMED	NO	NO	NO
4092069	Malignant neoplasm of sphincter of Oddi	187786003	SNOMED	NO	NO	NO
4116243	Malignant polyp of biliary tract	255087006	SNOMED	NO	NO	NO
4178966	Malignant tumor of ampulla of Vater	363417006	SNOMED	NO	NO	NO
4177105	Malignant tumor of extrahepatic bile duct	363416002	SNOMED	NO	NO	NO
4181327	Malignant tumor of gallbladder	363353009	SNOMED	NO	NO	NO
198695	Overlapping malignant neoplasm of biliary tract	109847004	SNOMED	NO	NO	NO
3655520	Perihilar cholangiocarcinoma	865951006	SNOMED	NO	NO	NO
36715928	Primary adenocarcinoma of ampulla of Vater	721718003	SNOMED	NO	NO	NO
37116592	Primary adenocarcinoma of biliary tract	733354000	SNOMED	NO	NO	NO
36717173	Primary adenocarcinoma of common bile duct	721559007	SNOMED	NO	NO	NO
36715798	Primary adenocarcinoma of cystic duct	721558004	SNOMED	NO	NO	NO
37018657	Primary adenocarcinoma of gallbladder	9541000119105	SNOMED	NO	NO	NO
36715927	Primary cholangiocarcinoma of intrahepatic biliary tract	721716004	SNOMED	NO	NO	NO
441225	Primary malignant neoplasm of ampulla of Vater	371967001	SNOMED	NO	NO	NO
194589	Primary malignant neoplasm of biliary tract	371970002	SNOMED	NO	NO	NO
4246034	Primary malignant neoplasm of common bile duct	93763008	SNOMED	NO	NO	NO
4247824	Primary malignant neoplasm of cystic duct	93770008	SNOMED	NO	NO	NO
40492416	Primary malignant neoplasm of distal bile duct	447416008	SNOMED	NO	NO	NO
40486896	Primary malignant neoplasm of extrahepatic bile duct	446189008	SNOMED	NO	NO	NO
197806	Primary malignant neoplasm of gallbladder	372139008	SNOMED	NO	NO	NO
40490929	Primary malignant neoplasm of intrahepatic bile duct	447109003	SNOMED	NO	NO	NO

40489795	Primary malignant neoplasm of perihilar bile duct	446807009	SNOMED	NO	NO	NO
37116589	Primary malignant neuroendocrine neoplasm of ampulla of Vater	733351008	SNOMED	NO	NO	NO
37116591	Primary malignant neuroendocrine neoplasm of biliary tract	733353006	SNOMED	NO	NO	NO
37116586	Primary malignant neuroendocrine neoplasm of cystic duct	733348001	SNOMED	NO	NO	NO
37116588	Primary malignant neuroendocrine neoplasm of distal bile duct	733350009	SNOMED	NO	NO	NO
42538835	Primary malignant neuroendocrine neoplasm of perihilar bile duct	762665002	SNOMED	NO	NO	NO
37116590	Primary mucinous cystic neoplasm with associated invasive carcinoma of biliary tract	733352001	SNOMED	NO	NO	NO
37118669	Primary mucinous cystic neoplasm with associated invasive carcinoma of cystic duct	733347006	SNOMED	NO	NO	NO
37116587	Primary mucinous cystic neoplasm with associated invasive carcinoma of distal bile duct	733349009	SNOMED	NO	NO	NO
37116585	Primary mucinous cystic neoplasm with associated invasive carcinoma of perihilar bile duct	733346002	SNOMED	NO	NO	NO
4311634	Secondary malignant neoplasm of ampulla of Vater	94164003	SNOMED	NO	NO	NO
4311638	Secondary malignant neoplasm of biliary tract	94185003	SNOMED	NO	NO	NO
4247946	Secondary malignant neoplasm of common bile duct	94262007	SNOMED	NO	NO	NO
4247947	Secondary malignant neoplasm of cystic duct	94270002	SNOMED	NO	NO	NO
4312301	Secondary malignant neoplasm of extrahepatic bile ducts	94291005	SNOMED	NO	NO	NO
4313079	Secondary malignant neoplasm of gallbladder	94312000	SNOMED	NO	NO	NO
4312024	Secondary malignant neoplasm of intrahepatic bile ducts	94349006	SNOMED	NO	NO	NO
35624313	Squamous cell carcinoma of gallbladder and extrahepatic biliary tract	766978002	SNOMED	NO	NO	NO
36674831	Squamous cell carcinoma of liver and intrahepatic biliary tract	770684008	SNOMED	NO	NO	NO
36674832	Undifferentiated carcinoma of liver and intrahepatic biliary tract	770685009	SNOMED	NO	NO	NO

A.2.7 Concept: [yiju] Plastic Biliary Stent

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
45771175	Polymeric biliary stent	701406003	SNOMED	NO	YES	NO
42087127	ADVANIX BILIARY STENT	J5202031	EDI	NO	NO	NO
42088559	BILIARY STENT	J5202105	EDI	NO	NO	NO
42089079	C-FLEX PIGTAIL BILIARY STENT	J5202021	EDI	NO	NO	NO
42090449	COTTON-LEUNG BILIARY STENT	J5202282	EDI	NO	NO	NO
42090450	COTTON-LEUNG BILIARY STENT	J5202382	EDI	NO	NO	NO
42092740	FLEXIMA BILIARY STENT SYSTEM	J5202011	EDI	NO	NO	NO
42098363	PERCUFLEX AMSTERDAM BILIARY STENT	J5202001	EDI	NO	NO	NO
42100653	SOLUS BILIARY STENT	J5203182	EDI	NO	NO	NO
42100842	ST-2 SOEHENDRA TANNENBAUM BILIARY STENT	J5203082	EDI	NO	NO	NO

42103466	ZIMMON BILIARY STENT	J5202482	EDI	NO	NO	NO
42103467	ZIMMON BILIARY STENT	J5202582	EDI	NO	NO	NO

A.2.8 Concept: [yiju] Biliary stent

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
4234567	Biliary stent	360123005	SNOMED	NO	YES	NO

B Outcome Cohort Definitions

B.1 Acute Cholangitis (Tokyo Guideline 2018, ATLAS Cohort ID: 697)

B.1.1 Cohort Entry Events

People enter the cohort when observing any of the following:

1. measurements of '[yiju] Body temperature', numeric value > 38; having at least 1 measurement of '[yiju] Total Bilirubin', starting between 0 days before and 0 days after '[yiju] Body temperature' start date; numeric value > 2.
2. measurements of '[yiju] WBC count', numeric value > 10; with all of the following criteria:
3. having at least 1 measurement of '[yiju] CRP', starting between 0 days before and 0 days after '[yiju] WBC count' start date; numeric value >= 1.
4. having at least 1 measurement of '[yiju] Total Bilirubin', starting between 0 days before and 0 days after '[yiju] WBC count' start date; numeric value > 2.

B.1.2 Cohort Exit

The person also exists the cohort at the end of continuous observation.

B.1.3 Cohort Eras

Remaining events will be combined into cohort eras if they are within 0 days of each other.

B.1.4 Concept: [yiju] Total Bilirubin

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
3024128	Bilirubin.total [Mass/volume] in Serum or Plasma	1975-2	LOINC	NO	NO	NO

B.1.5 Concept: [yiju] WBC count

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
3000905	Leukocytes [#/volume] in Blood by Automated count	6690-2	LOINC	NO	NO	NO

B.1.6 Concept: [yiju] Body temperature

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
3020891	Body temperature	8310-5	LOINC	NO	NO	NO

B.1.7 Concept: [yiju] CRP

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
3010156	C reactive protein [Mass/volume] in Serum or Plasma by High sensitivity method	30522-7	LOINC	NO	NO	NO

B.2 Moderate Acute Cholangitis (Tokyo Guideline 2018, ATLAS Cohort ID: 719)

B.2.1 Cohort Entry Events

People may enter the cohort when observing any of the following:

1. condition occurrences of '[yjpark] BS_Cholangitis'.

B.2.2 Additional Inclusion Criteria

I. hyperbilirubinemia Entry events having at least 1 measurement of '[yiju] Total Bilirubin', starting between 7 days before and 7 days after cohort entry start date; numeric value >= 5.

II. WBC Entry events having at least 1 measurement of '[yiju] WBC count', starting between 7 days before and 7 days after cohort entry start date; numeric value not Between 4 and 12.

B.2.3 Cohort Exit

The person also exists the cohort at the end of continuous observation.

B.2.4 Cohort Eras

Remaining events will be combined into cohort eras if they are within 0 days of each other.

B.2.5 Concept: [yiju] Hyperbilirubinemia

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
4032334	Hyperbilirubinemia	14783006	SNOMED	NO	YES	NO

B.2.6 Concept: [yiju] WBC count

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
3000905	Leukocytes [#/volume] in Blood by Automated count	6690-2	LOINC	NO	NO	NO

B.2.7 Concept: [yjpark] BS_Cholangitis

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
4267297	Acute cholangitis	6215006	SNOMED	NO	YES	NO

B.2.8 Concept: [yiju] Total Bilirubin

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
3024128	Bilirubin.total [Mass/volume] in Serum or Plasma	1975-2	LOINC	NO	NO	NO

B.3 Cholangitis - Broad Definition (ATLAS Cohort ID: 663)

B.3.1 Cohort Entry Events

People enter the cohort when observing any of the following:

1. condition occurrences of '[yiju] BS_Cholangitis'.
2. measurements of '[yiju] Total Bilirubin', numeric value > 10; having at least 1 measurement of '[yiju] WBC count', starting between 0 days before and 0 days after '[yiju] Total Bilirubin' start date; numeric value > 20.

B.3.2 Cohort Exit

The person also exists the cohort at the end of continuous observation.

B.3.3 Cohort Eras

Remaining events will be combined into cohort eras if they are within 0 days of each other.

B.3.4 Concept: [yiju] Total Bilirubin

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
3024128	Bilirubin.total [Mass/volume] in Serum or Plasma	1975-2	LOINC	NO	NO	NO

B.3.5 Concept: [yiju] WBC count

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
3000905	Leukocytes [#/volume] in Blood by Automated count	6690-2	LOINC	NO	NO	NO

B.3.6 Concept: [yiju] BS_Cholangitis

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
4267297	Acute cholangitis	6215006	SNOMED	NO	YES	NO

B.4 All-cause Mortality (ATLAS Cohort ID: 665)

B.4.1 Cohort Entry Events

People enter the cohort when observing any of the following:

1. death of any form.

B.4.2 Cohort Exit

The person also exists the cohort at the end of continuous observation.

B.4.3 Cohort Eras

Remaining events will be combined into cohort eras if they are within 0 days of each other.

B.5 Re-intervention (ATLAS Cohort ID: 696)

B.5.1 Cohort Entry Events

People may enter the cohort when observing any of the following:

1. device exposures of '[yiju] total biliary stent'.

2. procedure occurrences of '[yiju] reintervention procedure'.

3. procedure occurrences of '[yiju] ERCP, PTBD'.

B.5.2 Additional Inclusion Criteria

I. prior device exposure/ prior procedure Entry events with any of the following criteria:

1. having at least 1 device exposure of '[yiju] total biliary stent', starting anytime up to 7 days before cohort entry start date; allow events outside observation period.
2. having at least 1 procedure occurrence of '[yiju] ERCP, PTBD', starting anytime up to 7 days before cohort entry start date; allow events outside observation period.

B.5.3 Cohort Exit

The person also exists the cohort at the end of continuous observation.

B.5.4 Cohort Eras

Remaining events will be combined into cohort eras if they are within 0 days of each other.

B.5.5 Concept: [yiju] total biliary stent

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
45764662	Bare metal biliary stent	701398005	SNOMED	NO	YES	NO
45771175	Polymeric biliary stent	701406003	SNOMED	NO	YES	NO
37164911	Plastic biliary stent	1236977004	SNOMED	NO	NO	NO
42087127	ADVANIX BILIARY STENT	J5202031	EDI	NO	NO	NO
42088098	ARISTENT BILIARY CC	J5212024	EDI	NO	NO	NO
42088099	ARISTENT BILIARY CP	J5212124	EDI	NO	NO	NO
42088100	ARISTENT BILIARY NC	J8212024	EDI	NO	NO	NO
19693583	AURORA SELF-EXPANDING BILIARY STENT SYSTEM	J5212006	EDI	NO	NO	NO
19693572	BILIARY STENT	J5202098	EDI	NO	NO	NO
19693575	BILIARY STENT	J5202198	EDI	NO	NO	NO
42088559	BILIARY STENT	J5202105	EDI	NO	NO	NO
42103700	Biliary stent	J5212619	EDI	NO	NO	NO
19693580	BILIARY STENT ULTRAFLEX DIAMOND	J5212001	EDI	NO	NO	NO
19693568	BILIARY STENT, COTTON-HUIBREGTSE	J5202002	EDI	NO	NO	NO
19693569	BILIARY STENT, COTTON-LEUNG	J5202012	EDI	NO	NO	NO
19693581	BILIARY ZA-STENT	J5212002	EDI	NO	NO	NO
19693585	BONASTENT FOR BILIARY	J5212023	EDI	NO	NO	NO
42088879	BONASTENT FOR BILIARY	J8212123	EDI	NO	NO	NO
42088880	BONASTENT FOR BILIARY COVERED	J8212223	EDI	NO	NO	NO

42088881	BONASTENT FOR BILIARY WITH RETRIEVAL LASSO	J8212023	EDI	NO	NO	NO
42089079	C-FLEX PIGTAIL BILIARY STENT	J5202021	EDI	NO	NO	NO
42090177	COMVI BILIARY STENT	J5212519	EDI	NO	NO	NO
19693803	COMVI BILIARY STENT[COMVI ?□]	J8212819	EDI	NO	NO	NO
19693584	CORDIS SMART NITINOL STENT	J5212013	EDI	NO	NO	NO
19693571	COTTON-HUIBREGTSE BILIARY STENT	J5202082	EDI	NO	NO	NO
19693574	COTTON-HUIBREGTSE BILIARY STENT	J5202182	EDI	NO	NO	NO
42090449	COTTON-LEUNG BILIARY STENT	J5202282	EDI	NO	NO	NO
42090450	COTTON-LEUNG BILIARY STENT	J5202382	EDI	NO	NO	NO
19693576	COTTON-LEUNG SOF-FLEX BILIARY STENT	J5202682	EDI	NO	NO	NO
42090456	COVERED BILIARY	J5212004	EDI	NO	NO	NO
19693809	COVERED BILIARY WITH LASSO	J8212924	EDI	NO	NO	NO
42090460	COVERED LASSO BILIARY	J8212005	EDI	NO	NO	NO
42091505	EGIS BILIARY CONTRA-LATERAL LIMB STENT	J5212490	EDI	NO	NO	NO
42091506	EGIS BILIARY KEY-MB STENT, EGIS USAB STENT	J5212590	EDI	NO	NO	NO
42091507	EGIS BILIARY M-VALVE STENT, EGIS BILIARY WIDE STENT	J5212390	EDI	NO	NO	NO
42091512	EGIS DB BILIARY STENT	J5212090	EDI	NO	NO	NO
42091513	EGIS DC BILIARY STENT , EGIS BILIARY PARTIAL DC 20 STENT	J5212190	EDI	NO	NO	NO
42091518	EGIS FLOWER BILIARY BARE STENT	J5212790	EDI	NO	NO	NO
42091519	EGIS FLOWER BILIARY FULL COVER STENT, EGIS BILIARY FULLY COVERED STENT, EGIS BILIARY FLARED FULLY COVERED STENT, EGIS BILIARY FLARED FULLY COVERED VALVE STENT	J5212363	EDI	NO	NO	NO
42091520	EGIS FLOWER BILIARY PARTIAL COVER STENT	J5212690	EDI	NO	NO	NO
42091525	EGIS SB BILIARY STENT, EGIS BILIARY MOVING STENT	J5212163	EDI	NO	NO	NO
42091526	EGIS SC BILIARY STENT, EGIS K-PAPILLA STENT	J5212263	EDI	NO	NO	NO
19693586	ENDOCOIL BILIARY ENDOPROSTHESIS	J5212027	EDI	NO	NO	NO
42091808	EPIC NITINOL BILIARY STENT SYSTEM	J5212061	EDI	NO	NO	NO
42092740	FLEXIMA BILIARY STENT SYSTEM	J5202011	EDI	NO	NO	NO
19693588	FUSION ZILVER BILIARY STENT	J5212082	EDI	NO	NO	NO
42092969	GATE BILIARY COVERED STENT	J8212037	EDI	NO	NO	NO
42092970	GATE BILIARY UNCOVERED STENT	J8233037	EDI	NO	NO	NO
42093668	HANARO BILIARY SPIRAL STENT	J8212004	EDI	NO	NO	NO
42093888	HERCULES DH BILIARY	J5212063	EDI	NO	NO	NO
42093962	HILZO BILIARY COVERED STENT	J8212050	EDI	NO	NO	NO
42093963	HILZO BILIARY UNCOVERED STENT	J8212150	EDI	NO	NO	NO
19693582	JOSTENT SELFX(XF)	J5212003	EDI	NO	NO	NO
42095452	LUMINEXX3 BILIARY STENT	J5212068	EDI	NO	NO	NO
42097310	NEXENT BILIARY COVERED STENT	J5212196	EDI	NO	NO	NO
42097311	NEXENT BILIARY STENT	J8212096	EDI	NO	NO	NO
42097312	NEXENT BILIARY STENT	J8212196	EDI	NO	NO	NO
42097374	NITI-S BILIARY COVERED STENT	J5212119	EDI	NO	NO	NO
19693802	NITI-S BILIARY COVERED STENT [FLARE TYPE]	J8212719	EDI	NO	NO	NO
19693805	NITI-S BILIARY COVERED STENT[BOTH FLARE SONG SUTURE]	J8212920	EDI	NO	NO	NO
19693804	NITI-S BILIARY COVERED STENT[BUMPY SONG SUTURE]	J8212919	EDI	NO	NO	NO
19693806	NITI-S BILIARY COVERED STENT[BUMPY STRING]	J8212921	EDI	NO	NO	NO
42097375	NITI-S BILIARY COVERED STENT[BUMPY]	J5212419	EDI	NO	NO	NO

19693807	NITI-S BILIARY COVERED STENT[FULL COVERED SONG SUTURE]	J8212922	EDI	NO	NO	NO
19693808	NITI-S BILIARY COVERED STENT[HEAD TYPE]	J8212923	EDI	NO	NO	NO
42097376	NITI-S BILIARY COVERED STENT[KAFFES]	J8212120	EDI	NO	NO	NO
19693590	NITI-S BILIARY COVERED STENT[PD TYPE SONG SUTURE]	J5212791	EDI	NO	NO	NO
19693797	NITI-S BILIARY COVERED STENT[TWO-IN-ONE]	J8212019	EDI	NO	NO	NO
42097377	NITI-S BILIARY PTFE COVERED STENT(COMVI)	J5212319	EDI	NO	NO	NO
42097378	NITI-S BILIARY STENT	J5212019	EDI	NO	NO	NO
42097380	NITI-S BILIARY UNCOVERED STENT	J5212219	EDI	NO	NO	NO
42097422	NON COVERED BILIARY	J5212104	EDI	NO	NO	NO
19693798	NON COVERED LASSO BILIARY	J8212104	EDI	NO	NO	NO
19693772	OPTIMOS BILIARY PLASTIC STENT	J8202004	EDI	NO	NO	NO
19693800	PARTIAL COVERED BILIARY	J8212204	EDI	NO	NO	NO
42098363	PERCUFLEX AMSTERDAM BILIARY STENT	J5202001	EDI	NO	NO	NO
42099016	PROTEGE GPS SELF-EXPENDING NITINOL STENT BILIARY SYSTEM(SEBX-6-20-135 RX TYPE)	J5212073	EDI	NO	NO	NO
42099017	PROTEGE GPS SELF-EXPENDING NITINOL STENT BILIARY SYSTEM(SERB 65-06-20-80 OTW TYPE)	J5212173	EDI	NO	NO	NO
19693573	QUICK PLACE V	J5202101	EDI	NO	NO	NO
19693587	RX HERCULINK PLUS BILIARY STENT SYSTEM	J5212040	EDI	NO	NO	NO
42100041	SENTINOL NITINOL BILIARY STENT	J5212021	EDI	NO	NO	NO
42100653	SOLUS BILIARY STENT	J5203182	EDI	NO	NO	NO
42100793	SR SONG RETRIEVABLE BILIARY STENT	J5212028	EDI	NO	NO	NO
19693577	ST-2 SOEHENDRA TANNENBAUM BILIARY STENT	J5203002	EDI	NO	NO	NO
42100842	ST-2 SOEHENDRA TANNENBAUM BILIARY STENT	J5203082	EDI	NO	NO	NO
19693801	TARGETSECURE BILIARY STENT[COVERED]	J8212208	EDI	NO	NO	NO
19693799	TARGETSECURE BILIARY STENT[NON-COVERED, LASSO]	J8212108	EDI	NO	NO	NO
19693796	TARGETSECURE BILIARY STENT[NON-COVERED]	J8212008	EDI	NO	NO	NO
42103169	WALL STENT	J5212011	EDI	NO	NO	NO
42103171	WALLFLEX BILIARY STENT	J5212031	EDI	NO	NO	NO
42103172	WALLFLEX BILIARY STENT (UNCOVERED)	J5212041	EDI	NO	NO	NO
42103177	WALLSTENT ENDOPROSTHESIS WITH THE UNISTEP PLUS	J5212064	EDI	NO	NO	NO
42103458	ZILVER 635 BILIARY STENT	J5212202	EDI	NO	NO	NO
19693589	ZILVER BILIARY STENT	J5212102	EDI	NO	NO	NO
42103459	ZILVER BILIARY STENT	J5212051	EDI	NO	NO	NO
42103466	ZIMMON BILIARY STENT	J5202482	EDI	NO	NO	NO
42103467	ZIMMON BILIARY STENT	J5202582	EDI	NO	NO	NO
19693570	ZIMMON BILIARY STENT	J5202022	EDI	NO	NO	NO
4234567	Biliary stent	360123005	SNOMED	NO	YES	NO

B.5.6 Concept: [yiju] reintervention procedure

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
4238341	Removal of prosthesis of bile duct	58094005	SNOMED	NO	NO	NO
4280567	Repair of bile duct	366741003	SNOMED	NO	NO	NO

B.5.7 Concept: [yiju] ERCP, PTBD

Concept ID	Concept Name	Code	Vocabulary	Excluded	Descendants	Mapped
44789813	Percutaneous transhepatic biliary drainage single	223811000000102	SNOMED	NO	YES	NO
44791085	Percutaneous transhepatic biliary drainage multiple	223901000000107	SNOMED	NO	YES	NO
4201632	Percutaneous transhepatic insertion of biliary drain	315031003	SNOMED	NO	YES	NO

C Negative Control Concepts

Negative control outcomes are conditions or events with no plausible biological association with either metal or plastic biliary stent exposure. These are used for empirical calibration of systematic error.

Concept ID	Concept Name
4218106	Alcoholism
440424	Aphasia
378424	Astigmatism
261880	Atelectasis
80509	Bone cyst
434626	Borderline personality disorder
438407	Bulimia nervosa
134765	Cachexia
4172458	Candidiasis of skin
381581	Chalazion
4047787	Colles' fracture
198075	Condyloma acuminatum of the anogenital region
133228	Dental caries
433440	Dysthymia
376132	Ectropion

Concept ID	Concept Name
74855	Genital herpes simplex
195212	Hypercortisolism
140362	Hypoparathyroidism
4288544	Inguinal hernia
439840	Lymphangitis
436100	Narcolepsy
4215978	Onychomycosis
4171915	Orchitis
380731	Otitis externa
192606	Paraplegia
253796	Pneumothorax
4153877	Post-traumatic wound infection
373478	Presbyopia
194997	Prostatitis
380395	Retinal dystrophy
4339088	Testicular mass
440814	Torticollis
4028970	Tracheobronchitis
4114197	Tumor of hypothalamus
4092565	Uterine prolapse
197036	Vesicoureteric reflux
4223947	Viral hepatitis, type A
261326	Viral pneumonia
435613	Cellulitis
4049417	Vesicular eczema
436675	Anorexia nervosa
432590	Delusional disorder
194696	Dysmenorrhea
4135082	Dislocation of distal radioulnar joint
4001458	Fatigue fracture of vertebra
440129	Hypertrophy of nasal turbinates
380839	Tuberous sclerosis syndrome
134619	Toxic nodular goiter
377535	Sleep walking disorder
4239381	Psychoactive substance abuse
40443308	Polycystic ovary syndrome
376938	Neurofibromatosis syndrome
4304008	Memory impairment
374655	Mastoiditis
72418	Scoliosis deformity of spine
4142905	Fracture of rib

Concept ID	Concept Name
435093	Closed fracture of femur
42873169	Sprain of shoulder rotator cuff
316457	Mallory-Weiss syndrome
436676	Posttraumatic stress disorder
30234	Neck sprain
74189	Sprain of cruciate ligament of knee
139902	Allergic urticaria
376707	Acute conjunctivitis
437082	Ankylosing spondylitis
198803	Benign prostatic hyperplasia
377910	Deviated nasal septum
134453	Bursitis
138463	Aphthous ulcer of mouth
199861	Perianal abscess
4079750	Osteoarthritis of knee
4150372	Acute otitis media
141323	Acute maxillary sinusitis
433527	Endometriosis (clinical)
140168	Psoriasis
318800	Gastroesophageal reflux disease
81878	Benign paroxysmal positional vertigo
198202	Cystocele
374366	Sensorineural hearing loss
380094	Carpal tunnel syndrome
77965	Acquired trigger finger
72576	Benign tumor of breast
4207307	Infective meningitis
201792	Nongonococcal urethritis
4131595	Fracture of radius
4299544	Acanthosis nigricans
30441	Sialolithiasis
73300	Radial styloid tenosynovitis
135215	Hashimoto thyroiditis
4068482	Prolapsed lumbar intervertebral disc
433450	Paranoid schizophrenia
141933	Alopecia areata
381444	Contusion of eye
437116	Closed fracture of distal end of radius
4140300	Compression fracture of lumbar spine
4002998	Fracture of crown of tooth, enamel and dentin, without pulp exposure

Concept ID	Concept Name
4129394	Fracture of lumbar spine
4320022	Fracture of orbit
4134318	Fractured nasal bones
4069306	Idiopathic osteoporosis with pathological fracture
4069524	Osteoporotic fracture of lumbar vertebra
4133612	Tooth crown fracture
434500	Closed fracture of neck of femur
78272	Sprain of wrist
132466	Lumbar sprain
81151	Sprain of ankle
4133015	Sprain of anterior cruciate ligament of knee
40620346	Sprain of neck
74194	Sprain of spinal ligament
73649	Contusion of finger
4167102	Laceration of finger
436077	Developmental delay
4031019	Allergic contact dermatitis
134438	Contact dermatitis
133834	Atopic dermatitis
137053	Seborrheic dermatitis
4169287	Itching of skin
4300441	Burn scar
36713492	Non-erosive gastro-esophageal reflux disease
377558	Irregular astigmatism
380706	Regular astigmatism
75860	Constipation
132706	Disorder of nail
139099	Ingrowing nail
439776	Autism spectrum disorder